
PROJECT EMAIL SYSTEM (CLIENT)

Course: COMP.SEC.300 Secure Programming

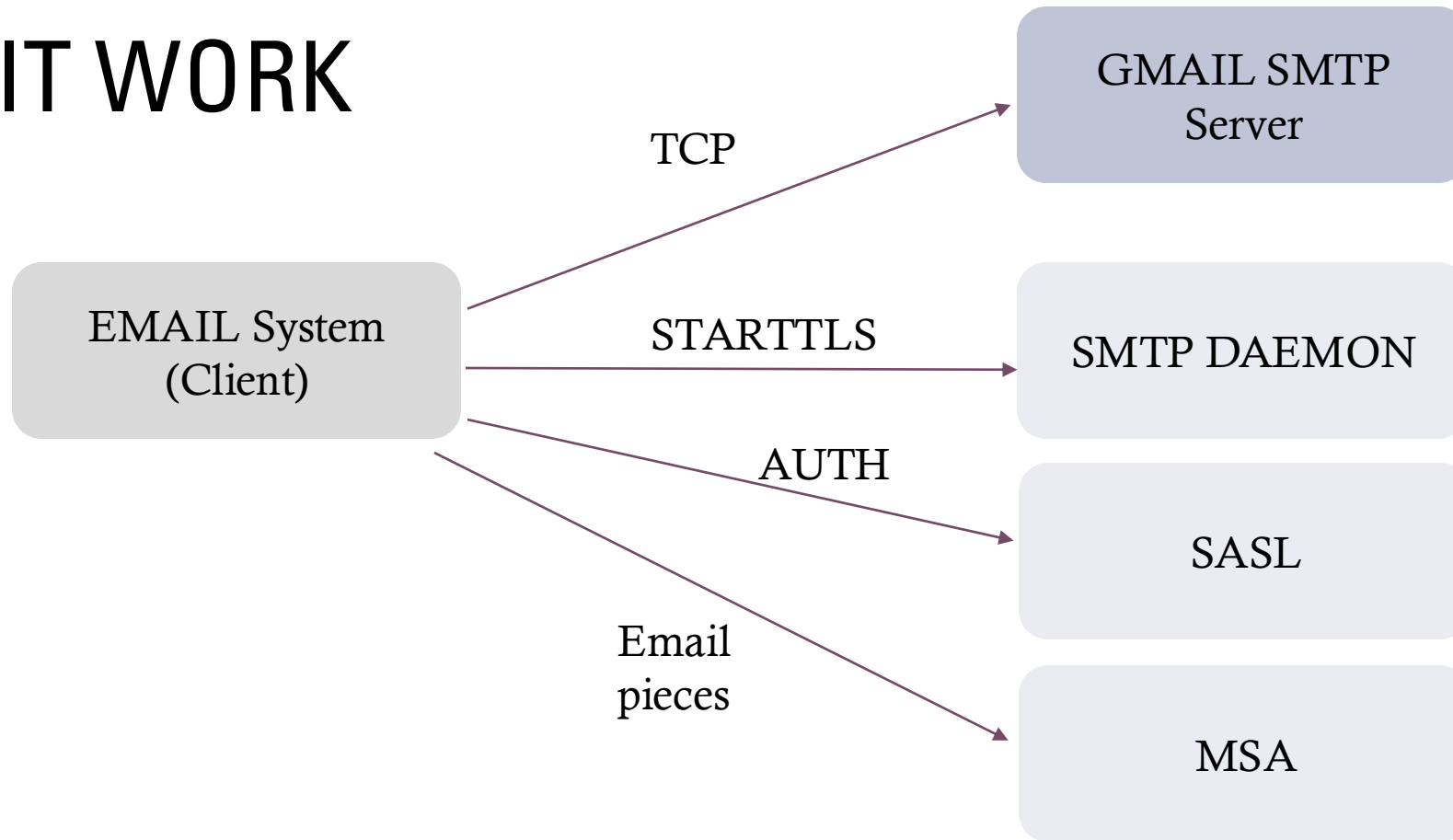
Student: Teemu Ranne

WHAT IS MY PROJECT

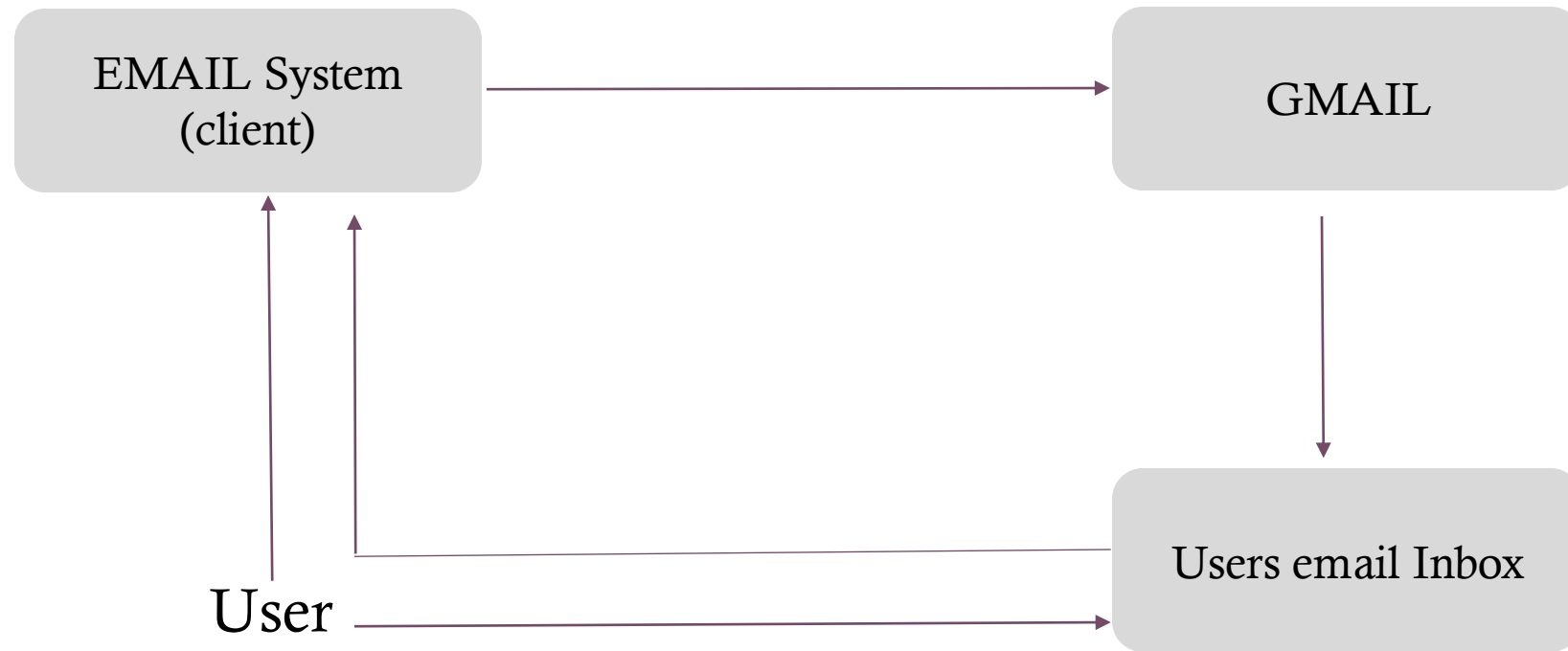
- Compiler: MinGW-w64 (GCC 11.2.0)
- Framework: Qt 6 (core and network modules)
- Package Manager (Windows): vcpkg configured with x64-mingw-static.
- Libraries:
 - Crypto++
 - nlohmann/json



HOW IT WORK



SECURITY AUTH



PROJECT DEMO TIME

OTHER SECURE PROGRAMMING SOLUTION

- DevSecOp pipeline
 - OWASP 10k worst password
 - SMTP Header Injection
 - Input validation
 - Email App Password (not permanently stored)
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SECURITY ISSUES AND VULNERABILITIES

- App user data stored in .txt
- nlohmann/json
- No time out for password attemps

FUTURE IMPLEMENTATION

- Server for the EMAIL System (client) app
 - SecureClear method
 - Recheck Snyk & Trivy
 - Log
 - Other features
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AI USAGE

- JsonConverter.cpp & .h
 - SaveAndLoad.cpp & .h
 - Hidden Input method & getch_wrapper
 - SMTP protocols AI was used extensively
 - Dedugging (Qt release build)
 - Initial information search
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